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pyufunc: A Set of Utility Functions that Keep Python Sweet

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Summary

pyufunc aims to bring together the most commonly used utility functions from different libraries and provide them in a single, cohesive package. By consolidating utility functions from multiple sources, pyufunc simplifies the process of finding and integrating various utility libraries into your projects. Whether you're a seasoned developer or just starting with Python, pyufunc provides a curated collection of utilities that cater to your everyday programming needs (Lott, 2018; Mertz, 2015).

Key Features

1. **Intuitive and Easy-to-Use:** Simplicity is at the core of pyufunc's design. Every utility function is thoughtfully documented, making it easy for developers of all skill levels to integrate them seamlessly into their projects.
2. **Modularity and Extensibility:** pyufunc is structured with modularity in mind. Each utility function is a standalone entity, allowing you to cherry-pick the ones you need without introducing unnecessary dependencies. Furthermore, the package is designed to be extensible, making it effortless to contribute your own utility functions and enrich the community.
3. **Robust Collection of Utility Functions:** pyufunc offers a versatile assortment of utility functions, carefully crafted and thoroughly tested to meet industry standards.
4. **Regular Updates and Maintenance:** Our team is dedicated to providing regular updates, ensuring that pyufunc remains compatible with the latest Python releases and industry best practices.
5. **Time and Effort Savings:** You can avoid reinventing the wheel by leveraging pre-existing, widely used utility functions. Let pyufunc take care of the repetitive tasks while you focus on building remarkable Python applications.

Dependencies won't be added to your environment unless you invoke functions that require them; when you do, the necessary packages are installed automatically.

If you identify functions that would benefit the broader community or have ideas for new utilities, please share your comments here: Issues or pull the repository and commit functions.

Statement of need

In the Python development community, efficiency and productivity are often hindered by repetitive tasks and scattered utility functions across numerous libraries. Developers frequently encounter the inconvenience of integrating multiple packages, each providing different subsets of common utility functions, leading to complexity in dependency management and integration challenges (Lott, 2018; Mertz, 2015).

pyufunc addresses this issue by consolidating the most frequently used utility functions into a single cohesive Python package. This centralized approach streamlines the coding experience, significantly reducing the overhead involved in identifying, installing, and managing various disparate utility libraries.

The necessity for a package like pyufunc arises from the following common scenarios faced by Python developers:

- **Fragmentation of utility functions**: Often, commonly used functions such as data manipulation, file handling, and mathematical operations are scattered across multiple libraries, each with its own dependencies, documentation standards, and installation processes.
- **Complexity in dependency management**: Integrating multiple small libraries often results in dependency conflicts or bloated virtual environments, making project setups cumbersome.
- **Repetitive development tasks**: Writing similar utility functions repeatedly in different projects consumes unnecessary time and resources.

By providing an intuitive, modular, and carefully maintained library, pyufunc simplifies these challenges. It ensures that Python developers have immediate access to a versatile set of robust, well-documented, and regularly updated utility functions (DeGrandis & Valetto, 2009; Walsh et al., 2004).

In conclusion, pyufunc fulfills a clear and practical need within the Python developer community, promoting efficiency, reducing redundant efforts, and enabling developers to focus on the core functionalities of their applications.

Existing Utility Functions Categorized by Functionality

The categories outlined in this document span a wide range of functionalities, each category is accompanied by a brief description, followed by a list of utility functions that fall under its umbrella, this comprehensive approach aims to arm developers with a robust toolkit, enabling them to select the most appropriate utility functions for their specific needs. Whether you are working on a complex application requiring advanced data manipulation or a simple project needing basic string operations, this guide endeavors to provide a valuable resource that enhances your development process and leads to more efficient, effective, and elegant coding solutions.

- utility_function_by_category.md

Existing Utility Functions Categorized by Keywords

By presenting these functions in a keyword-centric format, we facilitate a more intuitive and user-friendly approach to accessing a vast repository of tools, ensuring that developers can leverage the full potential of utility functions to optimize their code, improve performance, and innovate within their applications. Whether tackling complex algorithmic challenges or implementing basic functionality, this guide aims to be an essential companion, fostering a deeper understanding and more effective use of utility functions in software development projects.

- utility_function_by_keyword.md

Comprehensive Review of Existing Python Utility Function Packages

In this section, you will find a comprehensive review of existing utility function packages. Before delving into the evaluations and insights, the author wishes to express sincere gratitude to all the developers behind these packages. Your contributions to the open-source community are invaluable, and it is with great appreciation that we acknowledge your efforts and dedication.

Our goal with pyufunc is to collect all sorts of useful utility functions together to boost the efficiency of developers. If you need to use utility functions in your project, all you need is pyufunc.

Furthermore, we recognize the importance of proper usage and attribution of the utility functions we have integrated into pyufunc. If any package developer finds their utility function has been used improperly, we encourage you to reach out to the pyufunc developers for further discussion. We are committed to maintaining a respectful and collaborative relationship with the original developers, ensuring that all contributions are appropriately acknowledged and utilized within the bounds of open-source licenses and community norms. Your feedback and insights are crucial to us, as they help in refining pyufunc to better serve the open-source community.

Package Name	Description	Cite
pyutil	a library of useful Python functions and classes	tpltnt (2014)
PyHelpers	An open-source toolkit for facilitating Pythonusers' data manipulation tasks	Fu (2020)
psutil	Cross-platform lib for process and system monitoring in Python	Rodola (2009)
pyutilator	open source python package comprising of decorators that can be used for utility operations	Prince (2023)
pyutils	Python utilities	Gasch (2022)
common-pyutil	Bunch of common utility functions I've used in various projects.This package provides a uniform interface to them.	Badola (2019)
pyutl	functions and utilities to recycle code	Gómez (2019)
pyutilities	Useful utilities for python 3.10+	Dmitrii (2021)
dry-pyutils	This package's goal is to offer a set of utility methodsI end up using in a lot of projects.	Monteiro (2021)
pripy-utils	Python utilities	jaysen.lin (2025)
imutils	A series of convenience functions to make basic imageprocessing operations such as translation, rotation, resizing, skeletonization, and displaying Matplotlib images easier with OpenCV and Python.	PyImageSearch (2015)
dateutil	Useful extensions to the standard Python datetime features	dateutil (2015)
nb_utils	python utility functions	Boyane (2020)
Python-Charmers	A collection of useful python programs.	iwasaki (2020)
python-in-action	python crawler in action	PrinceCheng (2021)
tbm13-utils	Python utils made for personal use on my projects.	Como (2023)
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References

Badola, A. (2019). *Bunch of common utility functions i've used in various projects. This package provides a uniform interface to them*. GitHub. https://github.com/akshaybadola/common-pyutil

Boyane, N. (2020). *Python utility functions*. GitHub. https://github.com/Nivratti/nb_utils

Como, M. T. D. (2023). *Python utils made for personal use on my projects*. GitHub. https://github.com/TBM13/tbm13-utils

dateutil. (2015). *Useful extensions to the standard python datetime features*. GitHub. https://github.com/dateutil/dateutil

DeGrandis, P., & Valetto, G. (2009). Elicitation and utilization of application-level utility functions. *Proceedings of the 6th International Conference on Autonomic Computing*, 107–116.

Dmitrii. (2021). *Useful utilities for python 3.10+*. GitHub. https://github.com/dmitry-ed-gusev/pyutilities

Fu, Q. (2020). *PyHelpers: An open-source toolkit for facilitating python users' data manipulation tasks*. Zenodo. https://doi.org/10.5281/zenodo.4017438

Gasch, S. (2022). *Python utilities*. GitHub. https://github.com/scottgasch/pyutils

Gómez, J. (2019). *Functions and utilities to recycle code*. GitHub. https://github.com/Jesrat/pyutl

iwasaki, shuto. (2020). *A collection of useful python programs*. GitHub. https://github.com/iwasakishuto/Python-Charmers

jaysen.lin. (2025). *Python utilities*. GitHub. https://github.com/linjonh/pripy-utils

Lott, S. F. (2018). *Functional python programming: Discover the power of functional programming, generator functions, lazy evaluation, the built-in itertools library, and monads*. Packt Publishing Ltd.

Mertz, D. (2015). *Functional programming in python*. O'Reilly Media.

Monteiro, V. (2021). *A set of utility methods end up using in a lot of projects*. GitHub. https://github.com/monthero/dry-pyutils

Prince, A. (2023). *Open source python package comprising of decorators that can be used for utility operations*. GitHub. https://github.com/antoprince001/pyutilator

PrinceCheng. (2021). *Python crawler in action*. GitHub. https://github.com/Nevergiveupp/python-in-action

PyImageSearch. (2015). *A series of convenience functions to make basic image processing*

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operations such as translation, rotation, resizing, skeletonization, and displaying matplotlib images easier with OpenCV and python. GitHub. https://github.com/PyImageSearch/imutils

Rodola, G. (2009). *Cross-platform lib for process and system monitoring in python*. GitHub. https://github.com/giampaolo/psutil

tpltnt. (2014). *A library of useful python functions and classes*. GitHub. https://github.com/tpltnt/pyutil

Walsh, W. E., Tesauro, G., Kephart, J. O., & Das, R. (2004). Utility functions in autonomic systems. *International Conference on Autonomic Computing, 2004. Proceedings.*, 70–77.

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